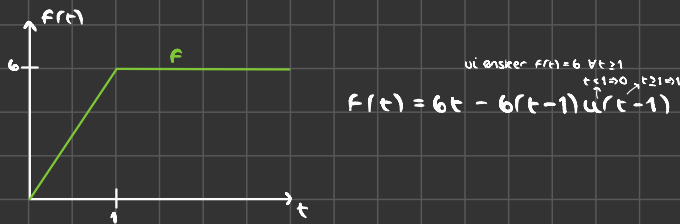


Prob 1 - Laplace transform [10 p]

a)
$$F(t) = \begin{cases} 6t, & 0 \leq t < 1 \\ 6, & t \geq 1 \end{cases}$$

We can rewrite the function using the Heaviside function.

Easier to see if we draw the function:



Then we just need to calculate the $\mathcal{L}t$ using the shift theorem

$$\mathcal{L}(F(t-a)u(t-a)) = e^{-sa}F(s)$$

$$F(s) = 6\mathcal{L}(t) - 6\mathcal{L}((t-1)u(t-1)) = \frac{6}{s^2} - \frac{6e^{-s}}{s^2}$$

b)
$$\mathcal{L}\left(\int_0^t \sin(x-t)y(x)dx\right) = \frac{Y(s)}{-s^2-1}$$

Since sine is an odd function, we can rewrite the left side as a convolution

$$\mathcal{L}\left(\int_0^t \sin(t-x)y(x)dx\right) = \frac{Y(s)}{-s^2-1} = -\frac{Y(s)}{(s^2+1)}$$

We have the theorem which says

$$\mathcal{L}(F * g) = \mathcal{L}\left(\int_0^t F(t-\tau)g(\tau)d\tau\right) = F(s)G(s)$$

This gives

$$\mathcal{L}\left(\int_0^t \sin(t-x)y(x)dx\right) = -\mathcal{L}(\sin(t)) \cdot \mathcal{L}(y(x)) = -\frac{1}{(s^2+1)} \cdot Y(s) = \frac{Y(s)}{-s^2-1}$$

Which is what we wanted to show.

c)
$$y(t) + \int_0^t \sin(x-t)y(x)dx = F(t)$$

$$Y(s) + \frac{Y(s)}{-s^2-1} = \frac{6}{s^2} - \frac{6e^{-s}}{s^2}$$

$$Y(s)\left(1 - \frac{1}{s^2+1}\right) = \frac{6}{s^2} - \frac{6e^{-s}}{s^2}$$

$$Y(s)\left(\frac{s^2}{s^2+1}\right) = \frac{6}{s^2} - \frac{6e^{-s}}{s^2}$$

$$\begin{aligned} Y(s) &= \frac{6(s^2+1)}{s^4} - \frac{6e^{-s}(s^2+1)}{s^4} \\ &= 6\left(\frac{1}{s^2} + \frac{1}{s^4} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s^4}\right) \end{aligned}$$

$$= 6(t + \frac{1}{6}t^3 - (t-1)u(t-1) - \frac{1}{6}(t-1)^3 u(t-1))$$

$$= \underline{6t + t^3 - 6(t-1)u(t-1) - (t-1)^3 u(t-1)}$$

Prob 2 - Fourier series [15p]

a) 1. $g_1(x) = f(ax) = f(ax + 2\pi) = f(a(x + \frac{2\pi}{a})) = g(x + \frac{2\pi}{a}) \Rightarrow P = \frac{2\pi}{a}$

2. $g_2(x) = f(x+a) \Rightarrow f(x+a+2\pi) = g(x+2\pi) \Rightarrow P = 2\pi$

3. $g_3(x) = f(x^2) \Rightarrow f(x^2+2\pi) \neq g(x+2\pi) = f((x+2\pi)^2)$

4. $g_4(x) = a + aF^a(x/a+a) \Rightarrow f(\frac{x}{a}+a+2\pi) = f(\frac{x+2\pi a}{a}+a) = g(x+2\pi a) \Rightarrow P = 2\pi a$

b) $f(x) = |\sin(x)|, x \in [-\pi, \pi]$

Sine is an odd function, but abs makes it even. Hence $f(x)$ is even. Then we know $b_n = 0$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} |\sin(x)| dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(x) dx \quad \begin{array}{l} f(x) \text{ is even: } \int_{-L}^L \rightarrow 2 \int_0^L, \sin(x) > 0 \forall x \in [0, L] \rightarrow \text{can remove abs.} \end{array}$$

$$= \frac{1}{\pi} [-\cos(x)]_0^{\pi}$$

$$= \frac{1}{\pi} (-\cos(\pi) - (-\cos(0))) = \frac{2}{\pi}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin(x)| \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(x+nx) + \sin(x-nx) dx$$

$$= -\frac{1}{\pi} \left[\frac{\cos(x(1+n))}{1+n} + \frac{\cos(x(1-n))}{1-n} \right]_0^{\pi}$$

$$\begin{array}{l} n \text{ does not } \rightarrow 0 \\ n \text{ per} \end{array} \quad = -\frac{1}{\pi} \left(\frac{\cos(\pi(1+n))}{1+n} + \frac{\cos(\pi(1-n))}{1-n} \right) - \left(\frac{1}{1+n} + \frac{1}{1-n} \right)$$

$$= -\frac{1}{\pi} \left(\frac{(-1)^{1+n} - 1}{1+n} + \frac{(-1)^{1-n} - 1}{1-n} \right)$$

$$= \frac{2}{\pi(1+n)} + \frac{2}{\pi(1-n)}$$

Prob 3 - Heat equation [12p]

$$u_t = c^2 u_{xx}, \quad x \in [0, \pi], \quad t \geq 0$$

$$u(0, t) = u(\pi, t) = 0, \quad t \geq 0$$

$$u(x, 0) = \frac{\pi}{2} - |x - \frac{\pi}{2}|, \quad x \in [0, \pi]$$

a) $u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} B_n \sin(nx) e^{-c^2 n^2 t}$

$$u(x, t) = F(x)G(t)$$

$$F(x)G'(t) = c^2 F''(x)G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{c^2 G(t)} = k$$

$$F''(x) - kF(x) = 0 \quad \wedge \quad G'(t) - kG(t) = 0$$

If $k=0 \Rightarrow F(x) = Ax + B$ but boundary cond. makes $A=B=0 \Rightarrow F(x)=0$ which is boring

If $k > 0 \Rightarrow F(x) = A e^{\sqrt{k}x} + B e^{-\sqrt{k}x} \quad \text{--- u --- since } e^{\sqrt{k}x} + e^{-\sqrt{k}x} \neq 0 \quad \forall x \in \mathbb{R}$

So $k < 0 \Rightarrow F(x) = A \cos(px) + B \sin(px)$ and $k = -p^2$

$$F''(x) + p^2 F(x) = 0 \quad \wedge \quad G'(t) + p^2 G(t) = 0$$

$$F(0) = A \cos(0) + B \sin(0) = 0 \Rightarrow A = 0$$

$$F(\pi) = B \sin(p\pi) = 0, \quad B \neq 0 \Rightarrow p = n$$

$$F_n(x) = B_n \sin(nx)$$

$$G'(t) + p^2 G(t) = 0 \Rightarrow G(t) = G_0 e^{-\lambda^2 t} = e^{-c^2 n^2 t}$$

$$u_n(x, t) = F_n(x)G_n(t) = B_n \sin(nx) e^{-c^2 n^2 t} \Rightarrow u(x, t) = \sum_{n=1}^{\infty} B_n \sin(nx) e^{-c^2 n^2 t}$$

which is what we wanted to show.

b)

$$b_n = \frac{1}{\pi} \int_0^{\pi} F(x) \sin\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi}{2} - |x - \frac{\pi}{2}|\right) \sin(nx) dx$$

$$= \int_0^{\pi/2} \sin(nx) dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (\pi - x) \sin(nx) dx$$

$$= \int_0^{\pi/2} \sin(nx) dx + 2 \int_{\pi/2}^{\pi} \sin(nx) dx - \frac{2}{\pi} \int_{\pi/2}^{\pi} x \sin(nx) dx$$

$$= \left[-\frac{\cos(nx)}{n} \right]_0^{\pi/2} + 2 \left[-\frac{\cos(nx)}{n} \right]_{\pi/2}^{\pi} - \frac{2}{\pi} \left[-\frac{x \cos(nx)}{n} + \int \frac{\cos(nx)}{n} dx \right]_{\pi/2}^{\pi}$$

$$= \left[-\frac{\cos(nx)}{n} \right]_0^{\pi/2} + 2 \left[-\frac{\cos(nx)}{n} \right]_{\pi/2}^{\pi} + \frac{2}{\pi} \left[\frac{x \cos(nx)}{n} - \frac{\sin(nx)}{n^2} \right]_{\pi/2}^{\pi}$$

$$\begin{aligned}
&= \frac{1 - \cos(\frac{n\pi}{2})}{n} + 2 \frac{\cos(\frac{n\pi}{2}) - \cos(n\pi)}{n} + \frac{2}{\pi} \left(\frac{\pi \cos(n\pi)}{n} - 0 - \left(\frac{\pi \cos(\frac{n\pi}{2})}{2n} - \frac{\sin(\frac{n\pi}{2})}{n^2} \right) \right) \\
&= \frac{1}{n} (1 - \cancel{\cos(\frac{n\pi}{2})} + 2\cancel{\cos(\frac{n\pi}{2})} - 2\cancel{\cos(n\pi)} + 2\cancel{\cos(n\pi)} - \cancel{\cos(\frac{n\pi}{2})} + \frac{\sin(\frac{n\pi}{2})}{n}) \\
&= \frac{n - \sin(\frac{n\pi}{2})}{n^2}
\end{aligned}$$

$$b_n = \begin{cases} \frac{1}{n}, & n \text{ jeun} \\ \frac{n - (-1)^n}{n^2}, & n \text{ odde} \end{cases}$$

$$u(x,t) = \sum_{n=1}^{\infty} \frac{1}{n} \sin(nx) e^{-c^2 n^2 t} + \sum_{k=0}^{\infty} \frac{2k+1 - (-1)^{2k+1}}{(2k+1)^2} \sin((2k+1)x) e^{-c^2 (2k+1)^2 t}$$

Prob 4 - Wave eq. [10p]

$$u_{tt} = \frac{1}{4} u_{xx}, \quad u(x,0) = x, \quad u_t(x,0) = \cos^2(x)$$

D'Alemberts solution:

$$\begin{aligned}
u(x,t) &= \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x} \\
&= \frac{1}{2} (x + \frac{1}{2}t + x - \frac{1}{2}t) + \int_{x-\frac{1}{2}t}^{x+\frac{1}{2}t} \cos^2(\tilde{x}) d\tilde{x} \\
&= x - \frac{1}{2} \int_{x-\frac{1}{2}t}^{x+\frac{1}{2}t} 1 - \cos(2\tilde{x}) d\tilde{x} \\
&= x - \frac{1}{2} \left[\tilde{x} - \frac{\sin(2\tilde{x})}{2} \right]_{x-\frac{1}{2}t}^{x+\frac{1}{2}t} \\
&= x - \frac{1}{2} (x + \frac{1}{2}t - \sin(2x+t) - (x - \frac{1}{2}t - \sin(2x-t))) \\
&= x - \frac{1}{2}t - \sin(2x+t) + \sin(2x-t) \\
&= x - \frac{1}{2}t - 2\sin(2x)\cos(t)
\end{aligned}$$

Prob 5 - Interpolation [9p]

x_i	-1	1	3
$f(x_i)$	1	4	3

a) $L_i = \prod_{j \neq i} \left(\frac{x - x_j}{x_i - x_j} \right)$

$$L_0 = \left(\frac{x - 1}{-1 - 1} \right) \left(\frac{x - 3}{-1 - 3} \right) = \frac{(x-1)(x-3)}{-2(-4)} = \frac{x^2}{8} - \frac{x}{2} + \frac{3}{8}$$

$$L_1 = \left(\frac{x - (-1)}{1 - (-1)} \right) \left(\frac{x - 3}{1 - 3} \right) = \frac{(x+1)(x-3)}{2 \cdot (-2)} = -\frac{x^2}{4} + \frac{x}{2} + \frac{3}{4}$$

$$L_2 = \left(\frac{x - (-1)}{3 - (-1)} \right) \left(\frac{x - 1}{3 - 1} \right) = \frac{(x+1)(x-1)}{4 \cdot 2} = \frac{x^2}{8} - \frac{1}{8}$$

$$u(x) = \sum y_i L_i = \frac{x^2}{8} - \frac{x}{2} + \frac{3}{8} + 4 \left(-\frac{x^2}{4} + \frac{x}{2} + \frac{3}{4} \right) + 3 \left(\frac{x^2}{8} - \frac{1}{8} \right)$$

b) $y_0 = 1 \quad y_1 = 4 \quad y_2 = 3$

$$[y_0, y_1] = \frac{4-1}{1-(-1)} = \frac{3}{2} \quad [y_1, y_2] = \frac{3-4}{3-1} = -\frac{1}{2}$$

$$[y_0, y_1, y_2] = \frac{-\frac{1}{2} - \frac{3}{2}}{3-1} = -1$$

$$N(x) = 1 + \frac{3}{2}(x-(-1)) - (x-(-1))(x-1)$$

$$= 1 - \frac{3}{2}x + \frac{3}{2} - x^2 - 1$$

$$= x^2 - \frac{3x}{2} + \frac{3}{2}$$

$$b) \quad (x_3, f_3) = (4, 6) \Rightarrow y_3 = 6$$

$$[y_2, y_3] = \frac{6-3}{4-3} = 3$$

$$[y_1, y_2, y_3] = \frac{3 - -\frac{1}{2}}{4-1} = \frac{7}{6}$$

$$[y_0, y_1, y_2, y_3] = \frac{\frac{7}{6} - 1}{4-1} = \frac{1}{3} \quad x^2 - 4x + 3$$

$$N(x) = x^2 - \frac{3x}{2} + \frac{3}{2} + \frac{1}{3}(x-(-1))(x-1)(x-3)$$

$$= x^2 - \frac{3x}{2} + \frac{3}{2} + \frac{x^3}{3} - \frac{4x^2}{3} + 1$$

$$= \frac{x^3}{3} - \frac{x^2}{3} - \frac{3x^2}{2} + \frac{5}{2}$$

Prob 7 - Nonlinear eq [8p]

$$x + \sin(x-2) = 0, \quad x \in [0, 2]$$

$$f(a) < k < f(b)$$

$$a < c < b$$

$$f(c) = k$$

$$f(0) = -0,9093$$

$$f(2) = 2$$

$$\exists c \in [0, 2] \text{ such that } f(c) = 0$$

$$f(x) = x + \sin(x-2)$$

$$f'(x) = 1 + \cos(x-2)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_1 = 0.5 - \frac{f(0.5)}{f'(0.5)} = 0,965$$

